INCH-POUND

MIL-B-8565/10(AS)
14 June 1993

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, HIGH-RATE, TYPE 2,
NICKEL-CADMIUM, 35 AMPERE-HOUR, CHARGE CONTROLLED

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-B-8565(AS)

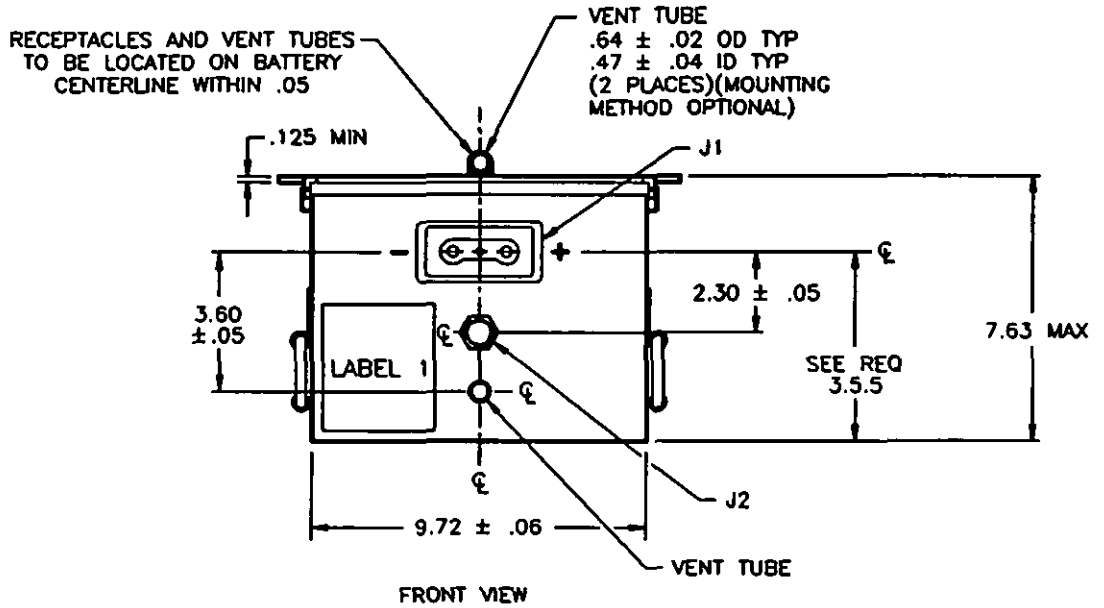
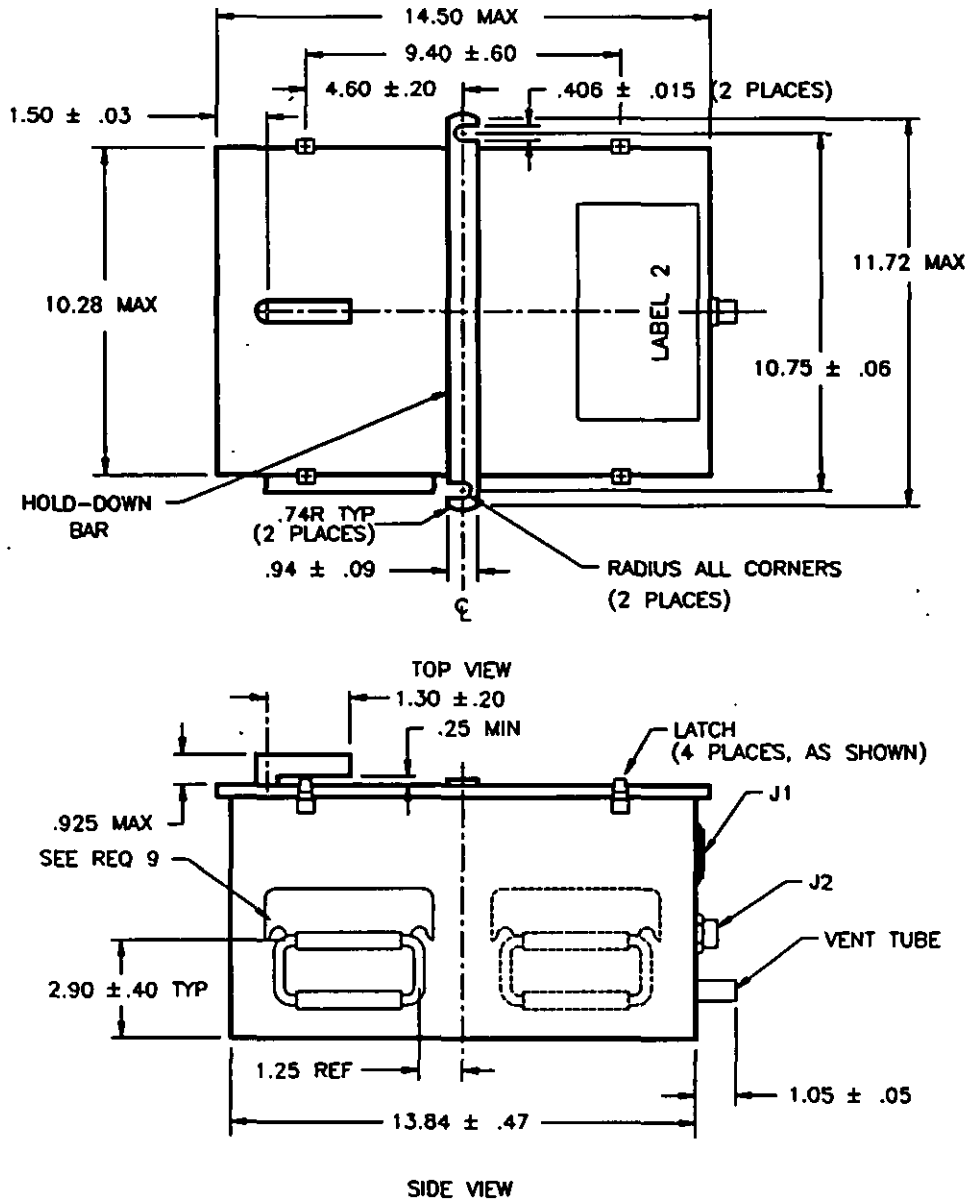


FIGURE 1.. PN D8565/10-1 dimensions.

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NOTES:

1. All dimensions are in inches.
2. Abbreviations used in this figure are inside diameter (ID), maximum (MAX), minimum (MIN), outside diameter (OD), part number (PN), requirement (REQ), and typical (TYP).

FIGURE 1. PN D8565/10-1 dimensions-Continued.

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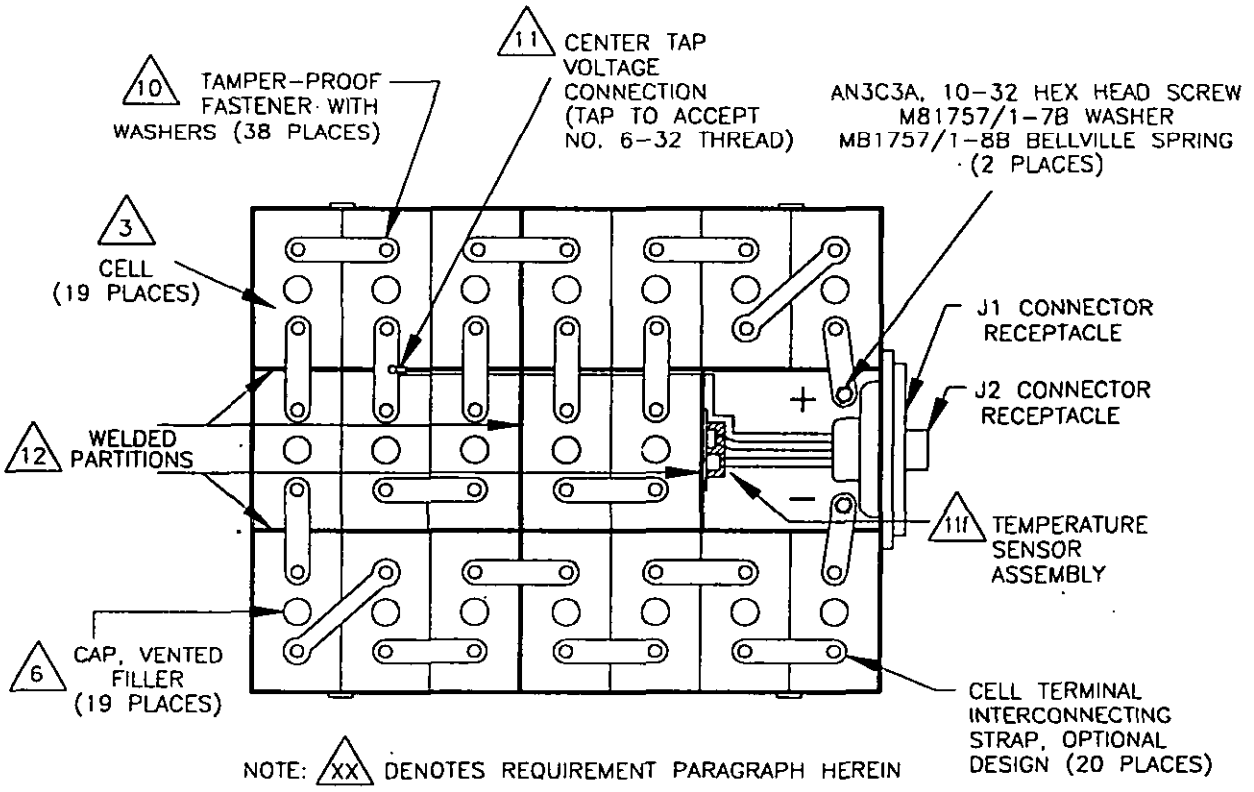


FIGURE 2. Layout of cells and sensors.

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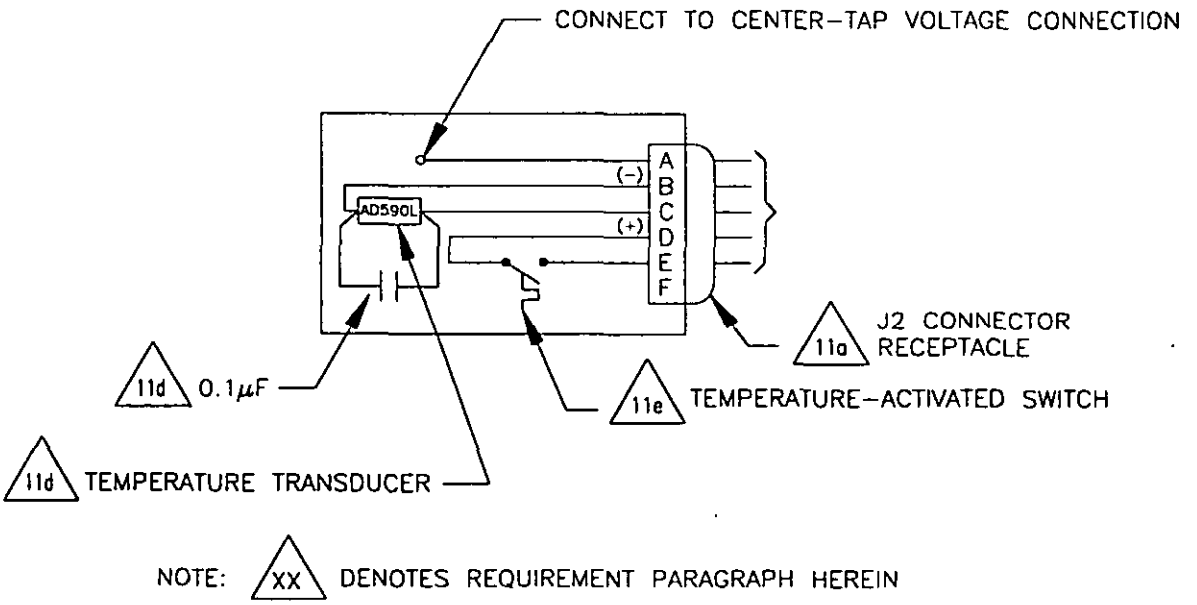
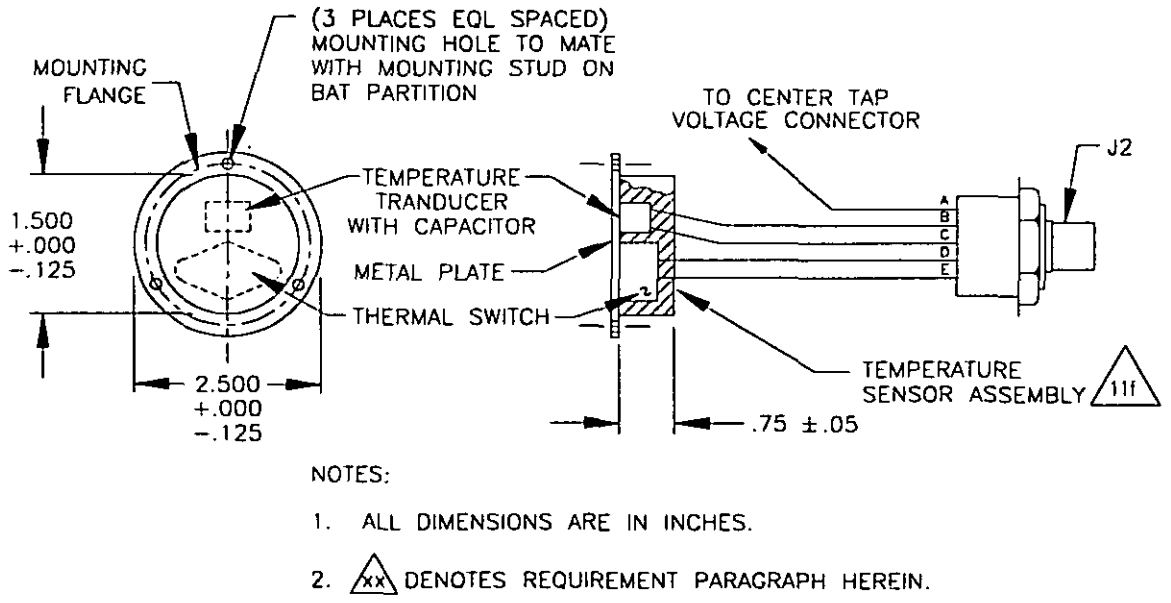


FIGURE 3. Electrical schematic of sensor/connector assembly.



NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. XX DENOTES REQUIREMENT PARAGRAPH HEREIN.

FIGURE 4. Removable sensor/connector assembly.

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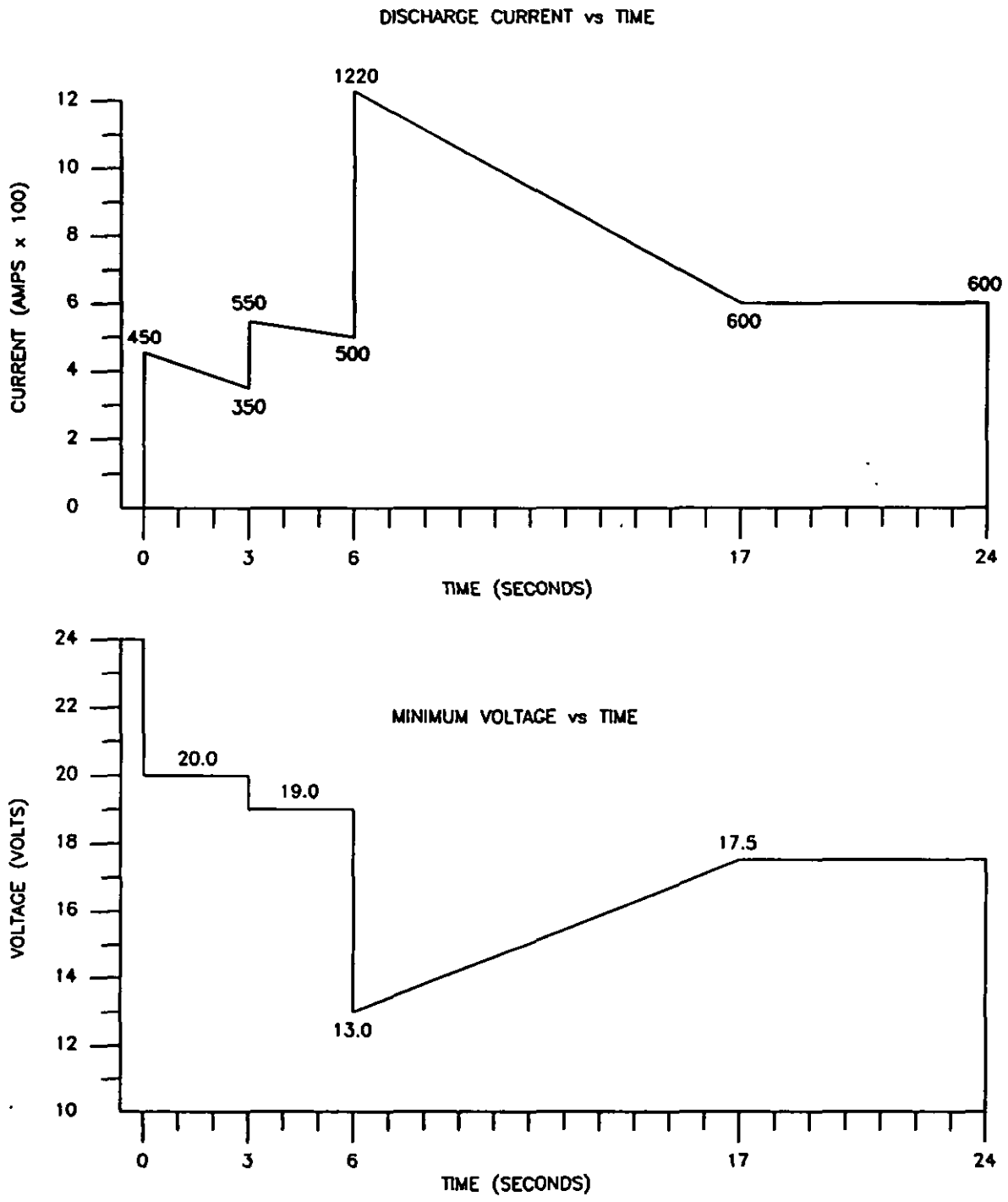
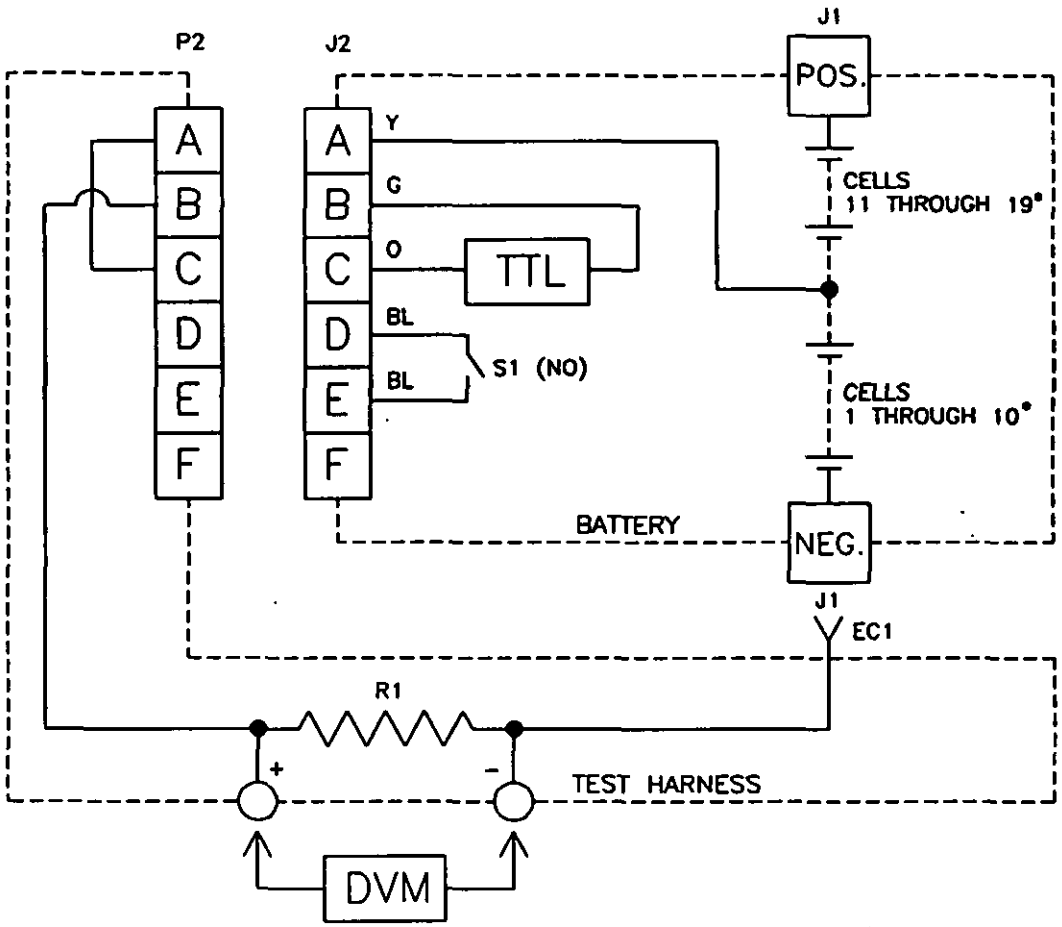


FIGURE 5. Profile discharge at -18°C.

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BATTERY

J1 – MS3509-28
J2 – CONNECTOR RECEPTACLE MS3474W10-6P
S1 – TEMPERATURE-ACTIVATED SWITCH:
CLOSES ON A TEMPERATURE RISE AT 74°C ±3°C
REOPENS ON A TEMPERATURE FALL DIFFERENTIAL OF 6°C MAXIMUM
TTL – TRANSISTOR-TRANSISTOR LOGIC DEVICE AD590L(4.0 TO 30.0 VOLTS DC)

TEST HARNESS

DVM – DIGITAL VOLTMETER
EC1 – ELECTRICAL CLIP W/INSULATOR
P2 – CONNECTOR PLUG, i.e. MS3476L10-6S (TO MATE WITH J2)
R1 – 10,000 OHM, 0.1 PERCENT, 1/10 WATT PRECISION RESISTOR

*NOTE: CELLS ARE NUMBERED NEGATIVE TO POSITIVE AS SCANNED BY THE NBC-1A CHARGER/ANALYZER
(REFERENCE TECHNICAL MANUAL NAVAIR17-15BAD-1)

FIGURE 6. Circuit for conducting sensor assembly function tests.

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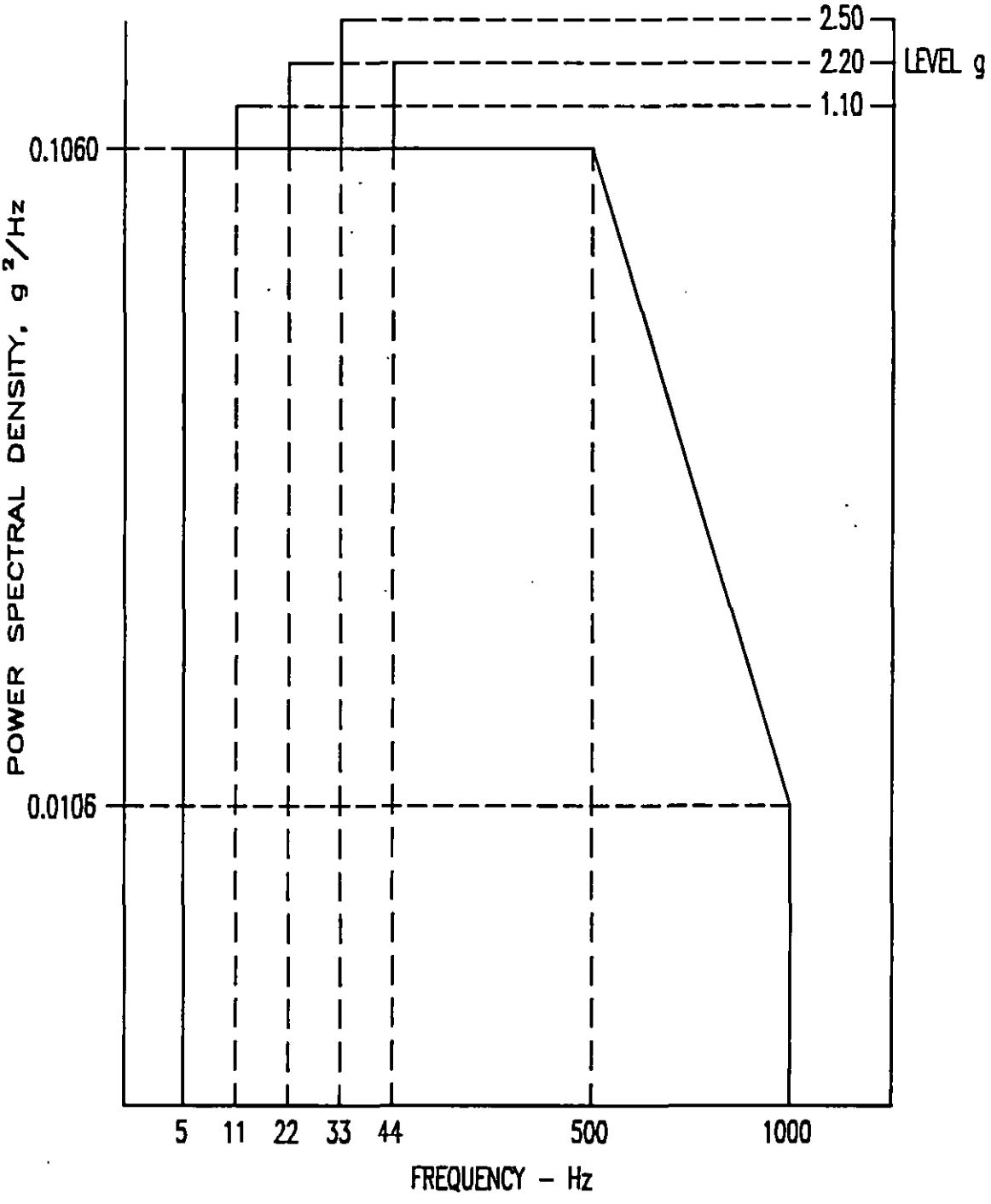


FIGURE 7. Vibration profile.

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REQUIREMENTS:

1. The part identification number of the battery shall be D8565/10-1.

2. The dimensions and configuration of the battery container shall be as shown on figure 1.

3. The cell layout of the battery shall conform to figure 2.

4. The 1C-rate capacity of the battery shall be 35 ampere-hours (Ah). The battery shall provide not less than 35 Ah capacity during the final discharge of the life cycle test based on a 35 ampere, constant current discharge for 1 hour, at 24°C ambient temperature, to a cutoff potential of 18.0 volts.

5. The weight of the battery shall be not greater than 85 pounds (38.64 kg).

6. The vented filler cap for each cell shall be MS3510-B.

7. If elastomers are used for cell post seals, the elastomers shall be made of ethylene propylene rubber or equivalent. Neoprene shall not be used.

8. The elastomeric gasket of the container cover shall have projections that contact each intercell connecting strap. The projections shall provide sufficient pressure to restrict vertical cell movement of the cells during vibration.

9. The container handles shall be part number 955-SS-SS-2-RG (manufactured by Nielsen Hardware, Zero Corporation, Hartford, CT 06141) or equal.

10. To impede cell removal, intercell connecting straps shall be attached to the cell posts by use of tamper-proof fasteners. Rivets shall not be used.

11. Each battery shall include a temperature sensor/connector assembly which shall be attached to the locations shown on figure 2. The temperature sensor/connector assembly shall include a temperature sensor assembly, a center tap voltage connection, the connector receptacle J2 and the interconnecting lead wires as shown on figures 3 and 4. This assembly shall not be damaged when tested, removed from the battery, in accordance with 4.6.20.3. When installed on the battery, this assembly shall be capable of withstanding without damage or degraded performance all of the environmental conditions to which the battery is subjected.

a. The J2 connector receptacle shall be in accordance with MIL-C-26482, Series II and shall mate with a connector plug conforming to MS3476L10-6S or equal. It shall be located as shown on figure 1 and electrically connected as shown on figure 3. The shell and contact pins shall be constructed of a nickel-alloy material such as inconel. The contact pins shall not be gold-plated. The connector receptacle need not be hermetically sealed.

b. The connecting lead wires shall be nickel plated copper, American Wire Gage (AWG) number 20 (19X32 stranding).

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c. The lead wire connections at the J2 receptacle shall be potted with an insulating compound MIL-P-47104(MI) or MIL-I-16923 (STYCAST 2651 or equal).

d. The temperature transducer shall be an Analog Devices, Inc. part number AD590L or equal. The capacitor shall be either an M39014/01-1473 or an M39014/02-1230 or equal.

e. The temperature-activated switch shall be a Texas Instruments, Inc. type M2S or equal. The switch contacts shall be normally open.

f. The temperature sensor assembly shall consist of the temperature transducer with capacitor and the temperature switch mounted to a metal plate with a potting cavity. The temperature sensor and the temperature activated switch shall be located near the center of the plate and potted inside the potting cavity. The exterior dimensions of the potting cavity shall be 1.500 +.000, -.125 inches in diameter and .75 ± .05 inches deep as shown on figure 4. The potting cavity shall be centered on the plate so that the plate forms a mounting flange as shown on figure 4. The plate shall be mounted to the partition which retains cell 16 from the negative terminal of the battery. The plate shall be mounted by means of the three holes shown on figure 4.

12. There shall be intimate contact between the partition that retains the center row of cells and the fourth cell from the positive terminal of the battery. No spacers or air passages shall be allowed at this interface. The partition shall be the same height as the plates within the cells that it retains, and it shall be rigid and strong enough to contain the cells for the life of the battery without warping or breaking. The partition shall have three, one-fourth inch long number 6-32 studs permanently affixed to it. The studs shall be equally spaced 120° ± 1° apart at the 12-, 4- and 8- o'clock positions on a 2.00 ± 05 inch diameter circle. The circle shall be centered on the face of the partition.

13. The required inspections to ensure the quality of the battery covered herein are listed in table I. The required inspections shall be conducted in the order listed, except the order of inspection numbers 2 through 11 may be interchanged.

14. The battery shall conform to MIL-B-8565J except as modified by the following:

14.1 Modify the following subparagraphs:

3.5.3 Latches (type 2 battery only). Following the last sentence, add: "The latches, strikes, and the outer seven-eighths inch of each end of the hold-down bar shall not be epoxy-coated."

3.5.4 Venting. Delete the third and sixth sentences.

3.5.5 Receptacles. Following the last sentence, add: "The J1 receptacle shall conform to MS3509-28. It shall be located so that the contact surfaces of the terminal block are in the same plane as the cell contact surfaces."

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3.5.6 Connectors, intercell. Delete the first sentence. Following the last sentence, add: "To impede cell removal, intercell connectors shall be attached to the cell posts by the use of tamper-proof fasteners. Rivets shall not be used. Epoxy or other plastics shall not be used to cover the intercell connectors or their fasteners."

3.5.7 Cells. Following the last sentence, add: "All cells shall be furnished in a wet, discharged, and fully formed state. The cells shall contain the proper amount of electrolyte to provide a level between one-eighth and one-fourth inch above the bottom of the baffle immediately after the end of charge in accordance with 4.5.1.4.2."

3.5.7.3 Cell terminals. Delete the first sentence.

3.5.8 Electrolyte leakage. Delete the last sentence.

3.5.10 Operating position. Delete "and 4.6.17" at the end of the sentence.

3.6.2 Venting (type 2 battery only). Delete "8 ± 2" and substitute: "6 ± 4".

3.6.3.1 Color. Following the last sentence, add: "If the battery case and lid are made of stainless steel, the battery may be identified as being an alkaline battery by having a 2-inch wide horizontal band of blue which crosses all external vertical surfaces of the battery container. The band shall be blue in accordance with table I."

3.6.3.2 Marking. Add the following subparagraph:

"3.6.3.2.6 Cell marking. Each cell top shall be impressed or embossed with a '+' marking, the manufacturer's name, production lot code, and the date manufactured. The '+' marking shall be placed near the positive (pos) terminal (term)."

3.6.3.2.2 Identification and warranty marking. Following the third sentence, add: "The first line of marking for label 1 shall be 'MIL PN: D8565/10-1'."

3.6.7 Vent tubes. N/A.

3.6.8 Capacity and electrical performance. Following the last sentence, add: "The battery shall have an initial capacity not less than 35 Ah when discharged in accordance with 4.6.10. The 1-minute constant voltage discharges shall be in accordance with 4.6.11. The current at 15 seconds and the average power for the 1-minute discharges shall be not less than:

Temperature	Average Power	Current at 15 Seconds
24°C	19.6 kW	1400 A
-30°C	9.1 kW	650 A

3.6.10.2c Life cycling (type 2 battery). Delete "85 percent" and substitute "100 percent".

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3.6.12d Environmental requirements. Delete the second sentence.

TABLE III. Battery environmental requirements. Delete "Mechanical shock" and "4.6.18".

3.6.15 Storage. N/A.

3.6.17 Humidity. Delete and substitute: "The battery shall meet the requirements of 3.6.12 and shall discharge for not less than 54 minutes before reaching cutoff voltage when subjected to the test of 4.6.24."

3.6.20 Gas barrier material (type 2 battery only). Delete and substitute: "The gas barrier material in each cell shall be intact and shall be removable in one complete piece when subjected to the test of 4.6.28."

14.2 Add the following paragraphs:

"3.6.23 Profile discharge at -18°C. The battery shall meet the minimum voltage requirements of figure 5 for each profile discharge when tested in accordance with 4.6.15.

"3.6.24 Handles. The handles shall show no evidence of breaking away from the case or being damaged when tested in accordance with 4.6.20.1. The method of mounting handles, latches, and strikes shall be optional.

"3.6.25 Vented filler cap seal. All venting shall occur only from the side vent band of the vented filler cap when tested in accordance with 4.6.20.2. Gas venting from any other location shall be a failure.

"3.6.26 Sensor assembly.

"a. The sensor assembly (including harness) shall not be adversely affected when immersed in 1.3 specific gravity potassium hydroxide electrolyte at 69°C for 168 hours. This shall be determined by conducting test 4.6.20.3.

"b. The accuracy of the temperature sensor assembly shall be ± 5 percent as determined from conducting test 4.6.20.4a.

"c. When subjected to 4.6.20.4b, the temperature switch contacts shall close at 74 ± 3°C on a temperature rise and reopen during a temperature drop. The temperature differential between closure and reopening of the contacts shall be not greater than 6°C.

"3.6.27 Switch contact resistance. When the contacts of the temperature-activated switch are in the open position, the resistance between the contacts shall be not less than 1.0 megohm as determined by conducting 4.6.20.5."

14.3 Modify the following subparagraphs:

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4.5.1.3 Constant potential charging. Delete "28.25 ± 0.50 volts" and substitute "28.25 ± 0.10 volts". Following the first sentence, add: "The voltage may fall outside this range for the first 5 minutes."

4.6.7 Strength of vent tubes. N/A.

4.6.10 Capacity discharge. Delete and substitute: "The battery shall be subjected to a 35-ampere discharge to an end voltage of 18.0 volts. The battery shall meet the initial capacity requirements of 3.6.8."

4.6.14 Capacity discharge at 49°C (120°F). N/A.

4.6.15 Charge and discharge test at low temperature. Delete and substitute: "The battery shall be subjected to the following in the sequence shown. The battery shall meet the requirements of 3.6.23 for every profile discharge conducted.

"a. Charge the battery in accordance with 4.5.1.4.2.

"b. Place the battery in a -18°C temperature chamber for 24 hours.

"c. Discharge the battery at 30 amperes for 20 minutes.

"d. Charge the battery at 28.25 volts for a period of 2 hours.

"e. Rest the battery on open circuit in a -18°C temperature chamber for 24 hours.

"f. Discharge the battery at 35 amperes for 2 minutes and then initiate the profile discharge of figure 5.

"g. After the profile discharge, rest the battery at 0.0 amperes for 4 minutes and then repeat the profile discharge of figure 5 one time.

"h. Repeat step g one time."

4.6.18 Mechanical shock. N/A.

4.6.20 Special tests. Add the following subparagraphs:

"4.6.20.1 Handle test. Each handle, when swung out from the case against its stop, shall have a force of 600 newtons (135 pounds) applied at the handle grasp and in a direction of maximum torque acting on the handle stop. The battery shall meet the requirements of 3.6.24.

"4.6.20.2 Vented filler cap seal test. Within 5 minutes after the end of the conditioning charge of 4.5.1.4.2, each battery shall be immersed in water. Observe where gasses are venting. The battery shall meet the requirements of 3.6.25.

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"4.6.20.3 Harness alkali resistance test. The sensor assembly harness shall be installed on the inside of a container which is capable of withstanding 1.3 specific gravity potassium hydroxide electrolyte. The harness connector shall be attached to the container in the same way it is attached to the battery. The container shall then be filled with electrolyte to cover all parts of the harness. The container shall then be placed in a 69°C environment to bring the electrolyte solution to 69°C. The harness shall soak in the electrolyte at 69°C for 168 hours. At the end of the 168-hour soak, the unit shall be examined for compliance with the physical requirements of 3.6.12 and 3.6.22. The unit shall meet the functional test requirements of 4.6.20.4 and 4.6.20.5.

"4.6.20.4 Sensor assembly function and switch activation tests. The electrical circuit shown on figure 6 shall be used to perform these tests. When conducting the sensor accuracy test, connect connector P2 to connector J2 and connect the insulated electrical clip EC1 to the negative terminal of the charged battery.

"a. Sensor accuracy test. Determine the ambient Fahrenheit temperature of the sensor as installed on the battery. This may be accomplished with the use of a calibrated glass stem thermometer. While the sensor is at this temperature, measure the voltage differential across resistor (R1) with the use of a calibrated digital voltmeter (DVM) as shown on figure 6. This measured differential voltage (ΔV) is to be applied in a formula to calculate a temperature in degrees Fahrenheit (T°F). The formula for this calculation is T°F = (ΔV - 2.554) X 180 [e.g., 77.8°F = (2.982 volts - 2.554) X 180]. The calculated temperature (T°F) shall be compared with the determined sensor temperature. The results shall comply with 3.6.26b.

"b. Sensor switch activation test. The temperature assembly shall have complied with 3.6.26b prior to performing this test. The temperature sensor shall be subjected to a temperature rise of 77°C maximum, and the temperature at which the switch closes shall be determined. When at this raised temperature and the switch closed, the temperature sensor shall be subjected to a temperature fall. The differential temperature at which the switch reopens shall be determined. As determined from the results of this test, the sensor switch shall comply with the requirements of 3.6.26c.

"4.6.20.5 Switch contact resistance test. The temperature-activated switch contacts shall be in the open position when performing this test. Measure the resistance between pins D and E of connector J2 shown on figure 3. The measured resistance shall comply with 3.6.27."

4.6.21 Temperature rise and float. Delete and substitute: "The following steps shall be performed in the order listed. Failure of the battery to comply with the requirements of 3.6.13 shall be cause for rejecting the battery. When performing the following steps, all necessary voltages, currents, times, etc. shall be monitored, measured, and recorded to determine compliance with 3.6.13. Unscheduled water/electrolyte additions shall not be performed.

"a. Charge the battery in accordance with 4.5.1.3.

"b. Remove the battery cover and place the battery in a temperature chamber at 49 ± 2°C for 12 hours.

"c. Discharge the battery at 315 amperes for 5 minutes or to 14.0 volts, whichever occurs first.

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"d. Immediately after the 315-ampere discharge and with the battery still in the temperature chamber at 49 ± 2°C, subject the battery to a constant potential charge of 28.5 + 0.2, - 0.0 volts for 8 ± 0.1 hours.

"e. Adjust the battery cell electrolyte to a level between one-eighth and one-fourth inch above the bottom of the cell baffle by adding 49°C distilled water.

"f. Repeat step c and then immediately resume the constant potential charge of 28.5 + 0.2, - 0.0 volts for 16 ± 0.1 hours with the battery in the 49°C environment.

"g. Adjust the electrolyte as described in step e.

"h. Repeat step c and then immediately resume the constant potential charge of 28.5 + 0.2, -0.0 volts for 24 ± 0.1 hours with the battery in the 49°C environment.

"i. Subject the battery to the requirement number (1) discharge in table II.

"j. Examine the battery for any physical damage resulting from performing steps a through i above. The battery shall comply with physical requirements a through f of 3.6.12."

4.6.23 Vibration. Delete the text of 4.6.23c and substitute: "Type 2 battery only. Vibrate along each orthogonal axis (in the axis order shown on figure 1 of the general specification) in accordance with the test levels specified on figure 7 of the specification sheet. The battery shall be vibrated for a period of 2 hours along each axis."

4.6.24d Humidity. Delete and substitute: "Discharge the battery at 35 amperes to an end-of-discharge voltage of 18.0 volts. The battery shall meet the requirements of 3.6.17."

4.6.26 Ground storage. N/A.

4.6.27c Physical integrity at high temperature (85°C). Delete the requirement of lifting the battery by the vent tubes.

4.6.28 Gas barrier material test (type 2 battery only). Delete and substitute: "Two or more cells from the sample 2 battery shall have the gas barrier material examined for compliance with 3.6.20."

4.6.28a through 4.6.28j N/A.

4.6.30 Final examination. Line 2: Delete "which can be detected without dissecting the battery".

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TABLE I. Qualification and quality conformance inspections.

Inspection number	Inspections	Qualification inspection			Refer to MIL-B-8565(AS) and MIL-B-8565/10		Quality conformance inspection		
		Qualification sample number			Requirements paragraph	Method of inspection paragraph	Group A sample no.	Group B sample no.	
		1	2	3			100%	1	2
1	Visual and mechanical examination	x	x	x	3.5, 3.6.3 & 3.6.22	4.6.3	x	x	x
2	Venting	x	x	x	3.6.2	4.6.2	x	x	x
3	Dimensions & weight	x		x	3.6.5	4.6.4	x	x	
4	Dielectric strength	x	x	x	3.6.4	4.6.5	x	x	x
5	Container body to container cover seal	x	x	x	3.6.6	4.6.6	x	x	x
6	Strength of handles	x	x	x	3.6.24	4.6.20.1	x	x	x
7	Color and marking	x	x	x	3.6.3	4.6.8	x	x	x
8	Strength of receptacle	x		x	3.6.9	4.6.12	x	x	x
9	Vented filler cap seal	x	x	x	3.6.25	4.6.20.2		x	x
10	Sensor assembly functions	x	x	x	3.6.26b & 3.6.26c	4.6.20.4a & 4.6.20.4b	x	x	x
11	Switch contact resistance	x	x	x	3.6.27	4.6.20.5	x	x	x
12	Discharge while inverted		x		3.5.8 & 3.6.8	4.6.16		x	
13	Conditioning charge	x	x	x	4.5.1.4.2	4.6.9		x	x
14	Capacity discharge	x	x	x	3.6.8	4.6.10	x	x	x
15	Altitude	x		x	3.6.11 & 3.6.12	4.6.17		x	
16	Charge and discharge at low temperature			x	3.6.23	4.6.15		x	
17	Temperature shock		x		3.6.11 & 3.6.12	4.6.19			x
18	Vibration		x		3.6.11 & 3.6.12	4.6.23			x

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TABLE I. Qualification and quality conformance inspections-continued.

19	Humidity		x		3.6.17 & 3.6.12	4.6.24			x
20	Salt fog		x		3.6.18 & 3.6.12	4.6.25			
21	Temperature rise and float		x		3.6.12 & 3.6.13	4.6.21			x
22	Life cycling	x		x	3.6.10.2	4.6.13.2		x	
23	Constant voltage discharge	x	x		3.6.8	4.6.11	x	x	x
24	Physical integrity at 85°C	x			3.6.11 & 3.6.12	4.6.27		x	
25	Final examination	x	x	x	3.6.4	4.6.30	x	x	x
26	Gas barrier material test			x	3.6.20	4.6.28			
27	Shelf life	x	x		3.6.15.2	4.6.10			
28	Alkaline resistance of sensor assembly harness			x	3.6.12, 3.6.22 & 3.6.26	4.6.20.3 & 4.6.20.4		x	x
29	Impact resistance	1/	1/	1/	3.6.19	4.6.27.1			1/
30	Corrosion resistance	2/	2/	2/	3.6.1	4.6.1			

Notes: 1/ For non-metallic battery containers, two outer containers that are representative of the three qualification samples shall be submitted.
2/ Metal parts that are representative of the metal parts contained in the three qualification samples shall be submitted.

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(Project 6140-N665)

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